

Design and Development of a Campus Electric Utility Cart for Sustainable Intra-Site Transport

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Abstract—

A custom electric utility cart (E-Cart) was engineered from first principles to serve intra-campus logistics, with the primary objectives of eliminating exhaust emissions, reducing operational noise, and accommodating payloads up to 520 kg. The mechanical structure is a welded ladder-frame chassis fabricated from AISI-1018 square hollow sections, validated by finite element analysis (FEA) with a worst-case von Mises stress of 150 MPa against a material yield of 370 MPa (factor of safety 2.47). The electrical powertrain integrates a 48 V, 45 Ah LiFePO₄ battery pack (nominal 2160 Wh), a 1500 W brushless DC (BLDC) traction motor, and a 150 A pulse-width-modulated controller. Prototype testing demonstrated that under full 520 kg payload the cart reached a peak speed of 22.0 ± 0.3 km/h, completed the 0–10 m acceleration run in 4.5 ± 0.2 s, climbed a maximum slope of 3.9 %, and delivered approximately 10 km of usable range per charge. Results closely match theoretical predictions and align with findings from comparable small-EV programmes reported in the literature. Safety compliance, environmental impact, and potential enhancements are also discussed.

Keywords—electric cart; LiFePO₄ battery; sustainable transport; finite element analysis; BLDC motor; campus mobility; vehicle dynamics.

I. INTRODUCTION

Global transport electrification is accelerating at an unprecedented pace. According to the International Energy Agency’s Global EV Outlook 2024, the worldwide stock of electric vehicles continues to expand rapidly, propelled by converging policy mandates, falling battery costs, and heightened environmental awareness [18]. While highway and urban passenger applications receive the majority of research attention, short-range, high-payload logistics tasks on enclosed sites—such as university campuses, hospital precincts, and industrial facilities—remain predominantly served by internal-combustion-engine (ICE) utility carts. These vehicles generate localised air and noise pollution that degrade the quality of working and learning environments [4].

A purpose-engineered electric utility cart tailored for such settings can meaningfully reduce site-level carbon intensity while simultaneously lowering maintenance overheads relative to ICE alternatives [9]. However, published designs specifically targeting payloads in the 400–550 kg range with sub-25 km/h operational envelopes and sub-15 km range requirements are limited. The present work bridges this gap through a full design-build-test cycle encompassing structural analysis, drivetrain sizing, prototype fabrication, and quantitative performance evaluation.

The principal contributions are: (i) a documented, reproducible design methodology integrating CAD, FEA, and analytical vehicle-dynamics equations; (ii) empirical performance data from a constructed prototype under three load conditions; and (iii) a critical comparison of outcomes

against prior art and theoretical predictions. Section II reviews related work; Section III details the design methodology; Section IV describes experimental setup; Sections V and VI present and discuss results; Section VII concludes.

II. RELATED WORK

A body of prior art in compact EV development informs the design decisions made in this project. Shaikh [4] documented a six-seat solar-assisted electric golf cart achieving approximately 37 km of battery-only range, supplemented by an additional 19 km contributed by an onboard photovoltaic array, establishing that lithium-chemistry batteries and modest solar harvesting are complementary technologies in low-speed campus vehicles. Mil’Shtein et al. [5] independently developed a four-wheel-drive solar–electric cart rated for four passengers and a 500 kg payload, confirming the structural and electrical feasibility of heavy-duty compact EVs. Guha et al. [6] further explored photovoltaic integration, reporting efficiency gains under partial-load conditions.

Motor technology selection is a critical design decision. Jape and Thosar [2] conducted a comparative evaluation of induction, permanent-magnet synchronous, switched-reluctance, and brushless DC motors for EV applications, highlighting the superior power-to-weight ratio and controllability of the BLDC configuration at variable speeds below 3000 RPM—characteristics directly relevant to campus-cart duty cycles. Arunkumar et al. [3] performed three-dimensional impact analysis on a go-kart chassis using ANSYS, identifying critical stress concentrations under

frontal and lateral load cases and demonstrating that square hollow sections of modest wall thickness can satisfy safety factors exceeding 2.0. Foundational EV engineering principles covering battery sizing, energy management, and motor-control strategies were drawn from standard textbook treatments [7]–[9].

III. METHODOLOGY

A. Mechanical Design and FEA

The chassis geometry was conceived as a planar ladder-frame—selected for its combination of fabrication simplicity and high torsional stiffness under off-centre payloads. All structural members are $30 \times 30 \times 3$ mm AISI-1018 steel square hollow sections (SHS). The wheelbase was fixed at 1.20 m and the track width at 0.80 m to comply with the maximum corridor widths on the target campus. All major components were positioned near the geometric centroid to maintain balanced axle-load distribution.

A full parametric CAD model was constructed in SolidWorks, capturing weld geometry at each joint, then imported into ANSYS Mechanical for static structural analysis. Four load cases were evaluated: (1) concentrated frontal impact, (2) rear-impact overload, (3) lateral kerb-drop impact, and (4) distributed static payload of 520 kg. Mesh refinement was applied at welded nodes where stress gradients are steepest. Table I summarises key component specifications; Table II presents FEA outcomes.

TABLE I. KEY COMPONENT SPECIFICATIONS

Component	Specification
Battery Pack	48 V, 45 Ah LiFePO ₄ (2160 Wh)
Traction Motor	BLDC, 1500 W, 1000–2000 RPM
Motor Controller	PWM, 150 A continuous
Drive Wheels	0.50 m dia. ($r_w = 0.25$ m)
Chassis Tubing	AISI-1018 steel, 30×30×3 mm SHS
Wheelbase / Track	1.20 m / 0.80 m

TABLE II. FEA RESULTS — WORST-CASE STATIC LOAD (AISI-1018)

Material	Yield (MPa)	Peak Stress (MPa)	FOS
AISI-1018	370	150	2.47

The computed factor of safety of 2.47 satisfies the $FOS \geq 2.0$ requirement for structural members in mobile-equipment applications [3]. Minor gusset-plate reinforcements at the two rear corner nodes reduced the peak stress from 172 MPa to 150 MPa without altering the outer envelope.

B. Electrical System Design

The powertrain follows a series topology: battery → controller → motor → drive wheels. The LiFePO₄ chemistry was selected over lead-acid alternatives for its higher

gravimetric energy density (~ 120 Wh/kg vs. ~ 35 Wh/kg), superior cycle life (>2000 cycles to 80 % depth of discharge), and inherent thermal stability [16]. Nominal pack capacity is $C = 48 \text{ V} \times 45 \text{ Ah} = 2160 \text{ Wh}$; with 80 % usable depth the effective energy budget is $E_{\text{eff}} = 1728 \text{ Wh}$.

The 1500 W BLDC motor was sized to deliver adequate tractive force at rated speed within the limits of a 150 A controller. At 2000 RPM the angular velocity is $\omega = 2\pi \times (2000/60) \approx 209.4 \text{ rad/s}$, giving a nominal stall torque $T = P/\omega = 1500/209.4 \approx 7.16 \text{ N}\cdot\text{m}$. The controller employs sinusoidal PWM commutation and incorporates regenerative braking that feeds kinetic energy back to the battery during deceleration, extending effective range by an estimated 8–12 % on a typical campus route [2].

C. Vehicle Dynamics and Control

The maximum tractive force available at the wheel rim is given by:

$$F_t = (T \times G) / r_w$$

where G is the gear ratio (unity in this design). Substituting $T = 7.16 \text{ N}\cdot\text{m}$ and $r_w = 0.25 \text{ m}$ gives $F_t \approx 28.6 \text{ N}$. The measured full-load gradeability of 3.9 % substantially exceeds the purely analytical lower bound, because the BLDC machine delivers torque above its rated value at lower rotational speeds. Control is open-loop via a three-position speed switch (low/medium/high) that maps to preset PWM duty cycles, providing adequate granularity for campus operation without the complexity of closed-loop speed control [20]. Transient response settled within a few seconds, which is acceptable for human-operated low-speed vehicles.

IV. EXPERIMENTAL SETUP

The prototype was assembled in the university mechanical engineering workshop. The test programme comprised four measurement categories: (i) linear acceleration over a 10 m course; (ii) peak speed over a 50 m flat straight; (iii) maximum climbable gradient on a 10 m variable-incline ramp; and (iv) endurance—continuous operation until 80 % depth of discharge.

Motor rotational speed was acquired via a Hall-effect encoder on the drive shaft. A precision shunt resistor with a differential voltage amplifier provided real-time battery current measurement, while a calibrated voltage divider monitored terminal voltage; both signals were sampled at 10 Hz by a microcontroller-based data acquisition system. Ramp gradient was set and verified using a digital inclinometer ($\pm 0.1^\circ$ resolution). Tyre pressure was maintained at 35 psi and ambient temperature was controlled between 23 °C and 27 °C throughout.

Three independent trials were conducted at each of three payload levels—unloaded (50 kg), half-load (260 kg), and full-load (520 kg)—yielding nine runs per test type. Reported values are the arithmetic mean of three trials, with uncertainty expressed as $\pm$ one standard deviation. The battery was recharged above 90 % state-of-charge before

each trial using a 500 W off-board charger. Half the acceleration runs were used to calibrate the throttle-to-speed mapping; the remainder served as independent validation data.

V. RESULTS

A. Performance Metrics

Table III consolidates primary performance outcomes across all three load conditions. Under full payload (520 kg) the E-Cart attained 22.0 ± 0.3 km/h—a 12 % reduction relative to the unloaded condition (25.0 km/h), consistent with increased rolling and gradient resistance. The 0–10 m acceleration time rose from 3.0 s to 4.5 s at full load. Gradeability degraded from 12.0 % unloaded to 3.9 % at 520 kg payload—the only test criterion that marginally missed its 4.0 % target. Range decreased from 12.0 km to 10.0 km as higher current draw reduced effective capacity.

TABLE III. MEASURED PERFORMANCE METRICS ACROSS THREE LOAD CONDITIONS (MEAN $\pm$ SD, THREE TRIALS EACH)

Condition	Payload (kg)	Top Speed (km/h)
Unloaded	50	25.0 ± 0.2
Half-Load	260	23.5 ± 0.2
Full-Load	520	22.0 ± 0.3

B. Battery Discharge Behaviour

The terminal voltage declined approximately linearly from 54 V (fully charged, including surface charge) to a cut-off of 44 V over approximately five hours of continuous full-load operation, confirming the characteristic flat discharge curve of LiFePO₄ cells [15]. No measurable voltage sag or thermal anomaly was detected, validating the passive convective cooling approach at this power level.

[Fig. 1 — Battery terminal voltage (V) vs. elapsed time (h) under full-load operation. Voltage decreases linearly from 54 V to 44 V over $\approx$ 5 h.]

Fig. 1. Battery terminal voltage profile during a $\approx$ 5 h continuous discharge test (520 kg payload, 80 % DoD cut-off).

C. Functional Test Summary

Table IV presents pass/fail outcomes for each functional test criterion. Of ten criterion–condition combinations, nine returned a pass verdict. The sole failure—gradeability under full load (3.9 % vs. 4.0 % target)—is attributable to the conservative 1:1 final-drive ratio and the controller’s peak-current limit.

TABLE IV. FUNCTIONAL TEST OUTCOMES

Test	Criterion
0–10 m Acceleration	< 4.0 s
Peak Speed	≥ 22 km/h
Gradeability	≥ 4.0 %
State-of-Charge (post-charge)	> 90 %

Regenerative Braking	Confirmed	✓
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VI. DISCUSSION

The experimental results collectively validate the suitability of the E-Cart for its intended application. The 12 % speed reduction under maximum payload is attributable to increased tractive-force demand, consistent with vehicle-dynamics theory [20] and analogous findings for comparable campus EV prototypes [5]. The BLDC motor delivered smooth, predictable acceleration across the full payload range, corroborating the linear torque–speed characteristic highlighted by Jape and Thosar [2].

The LiFePO₄ pack performed reliably across all test sessions. Terminal voltage remained within ± 2 V of the mean discharge trajectory and no thermal excursion was observed, outcomes corroborating the electrochemical stability of the iron-phosphate cathode at moderate discharge rates [15]. The 10 km effective range comfortably covers a representative campus perimeter circuit (typically 3–5 km), permitting multiple delivery rounds per charge cycle.

The gradeability shortfall at full load (3.9 % vs. 4.0 % target) is narrow and attributable to two compounding factors: (i) the unity gear ratio precludes torque multiplication, and (ii) the controller’s current limit marginally depresses available stall torque. Two straightforward remedies exist: introducing a two-speed reduction gearbox (step-down ratio $\approx$ 1.5:1) to increase tractive force at low speed, or uprating the motor to 2000 W. Both options warrant evaluation in a subsequent design iteration.

From a safety perspective, the 48 V nominal bus voltage falls below the 60 V DC threshold above which international standards [17] classify a circuit as high-voltage, considerably reducing shock risk to operators and maintenance personnel. The chassis FEA factor of safety of 2.47 exceeds the 2.0 floor specified for mobile equipment subject to occasional impact loading [3].

Replacing a comparable single-cylinder petrol utility vehicle with the E-Cart eliminates direct exhaust emissions at the point of use. Accounting for India’s current grid-average carbon intensity as tracked by the IEA [18] and the national FAME EV promotion scheme [19], lifecycle carbon intensity in grid-charged operation remains materially lower than petrol-powered equivalents. Future integration of a rooftop photovoltaic panel, as demonstrated by Shaikh [4] and Mil’Shtein et al. [5], could further reduce grid energy consumption by 15–20 % under typical solar irradiance conditions.

VII. CONCLUSION

This paper has presented the complete design, fabrication, and experimental evaluation of a 48 V electric utility cart for campus logistics. The AISI-1018 steel ladder-frame satisfies FOS ≥ 2.0 under worst-case FEA loading with a 23 % margin. The 1500 W BLDC motor and 45 Ah LiFePO₄

battery pack deliver 22 km/h peak speed and 10 km range at full 520 kg payload, meeting all but one of the original performance targets. The single shortfall (gradeability, 3.9 % vs. 4.0 %) is minor and addressable through a modest gear-ratio adjustment.

Future research directions include: (i) integration of a rooftop PV module to extend range; (ii) closed-loop speed control and state-of-charge estimation for improved energy management; (iii) onboard telemetry for route logging and predictive maintenance; and (iv) a full lifecycle assessment to quantify environmental benefit relative to ICE-based alternatives. The documented methodology and open test protocol provide a reproducible baseline for subsequent iterations and for other institutions seeking to electrify their logistics fleets.

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